Angle bisectors and perpendicular bisectors

1 Complete the following statements:

a A point is on the perpendicular bisector of a line segment if and only if . . .

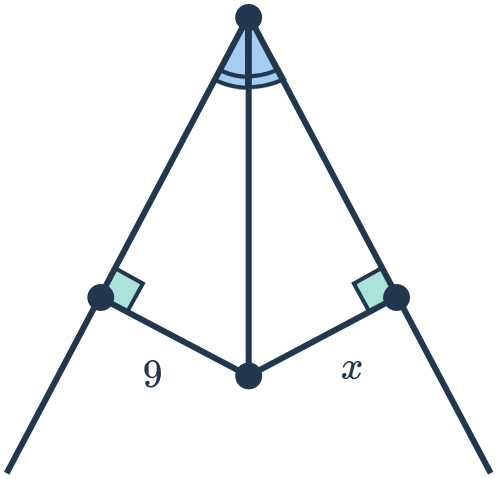
b A point is on the bisector of an angle if and only if . . .

2 For each of the following diagrams:

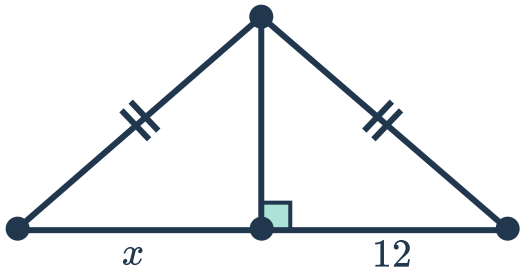
i Find x.

ii Justify your answer.

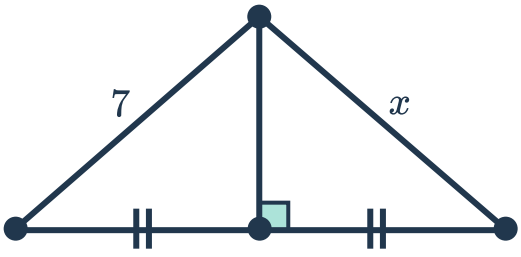
a



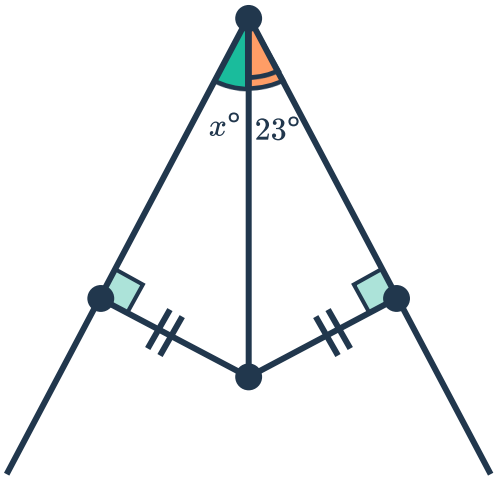
b



c



d



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

3 For each of the following triangle types, construct the perpendicular bisectors of the three sides:

a Acute triangle

b Right triangle

c Obtuse triangle

d Equilateral triangle

4 For each of the following triangle types, construct the angle bisectors of the three angles:

a Acute triangle

b Right triangle

c Obtuse triangle

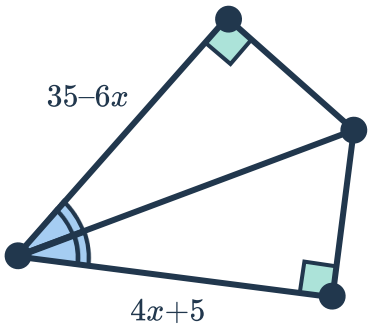
d Equilateral triangle

5 For each of the following diagrams:

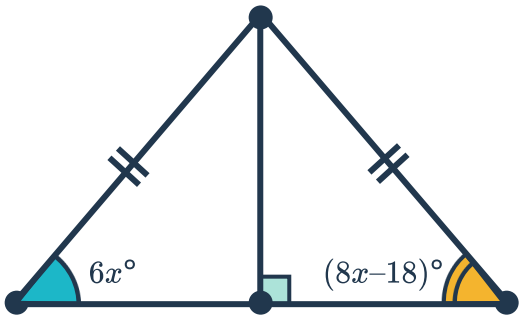
i State the theorem and congruence test used to prove the two triangles are congruent.

ii Find x.

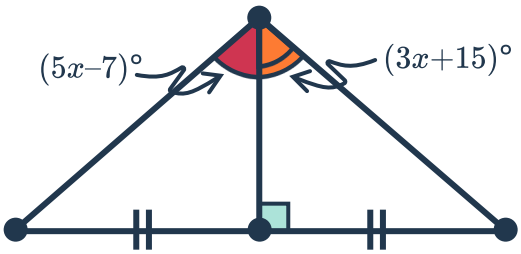
a



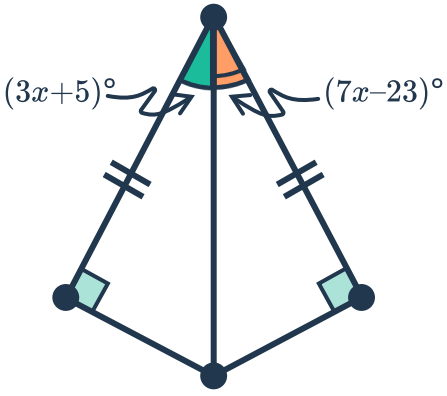
b



c



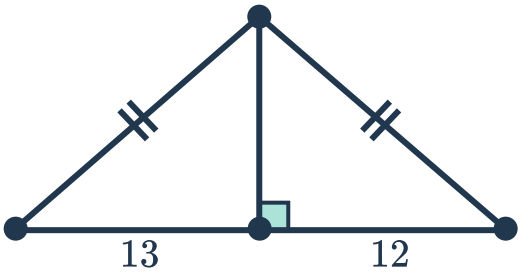
d



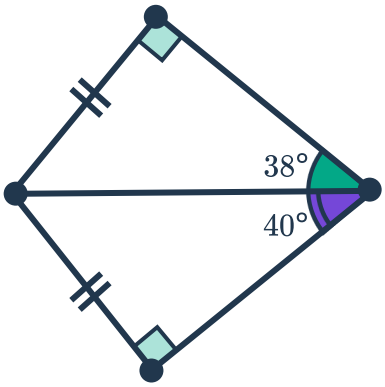
© ClipSAT: By Mr. Mohamed Abdallah +966597688647

6 Determine whether each of the following diagrams is valid. If not, identify the contradiction.

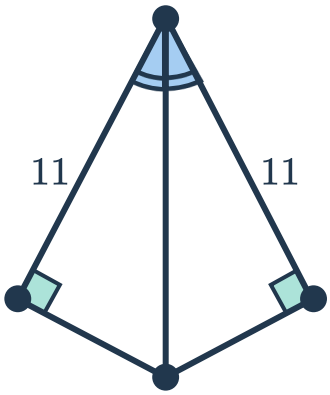
a



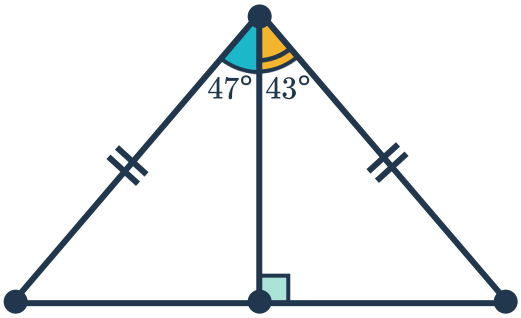
b



c



d

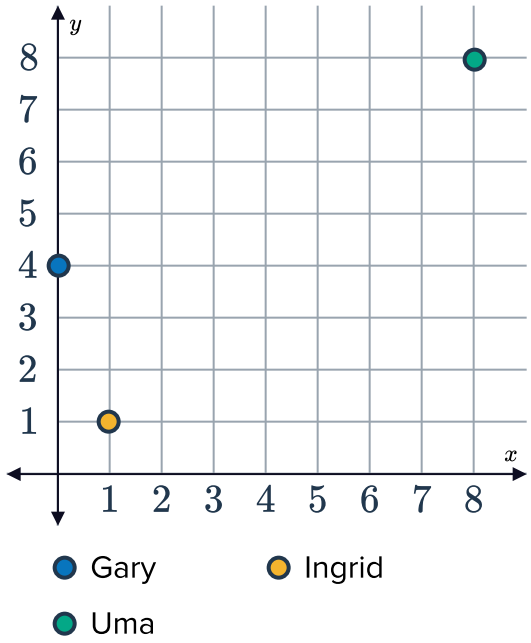


© ClipSAT: By Mr. Mohamed Abdallah +966597688647

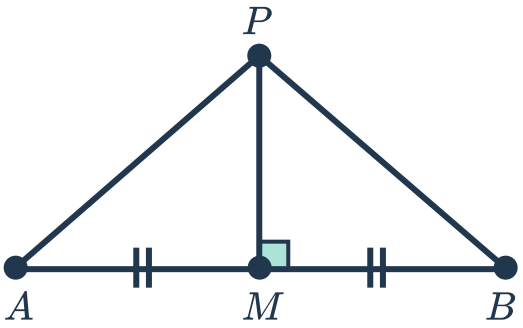
7 Gary, Uma and Ingrid are deciding on a location to meet up at during the holidays, such that the meeting place is equidistant from each of their homes.

a How can they find the exact coordinates for such a meeting place?

b Copy the diagram and make the appropriate constructions to find the meeting place.



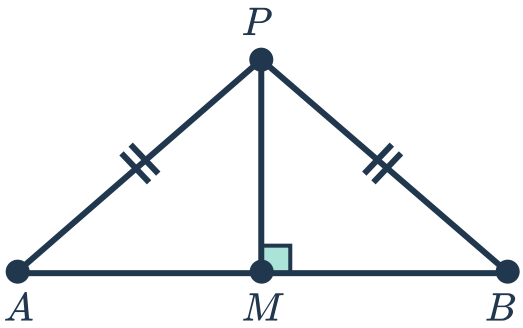
8 Construct a proof of the perpendicular bisector theorem.
Given: $\overline{PM}$ is the perpendicular bisector of $\overline{AB}$
Prove: $PA = PB$



9 Construct a proof of the converse of perpendicular bisector theorem.
Given:

- $PA = PB$
- $\overline{PM} \perp \overline{AB}$

Prove: $AM = BM$

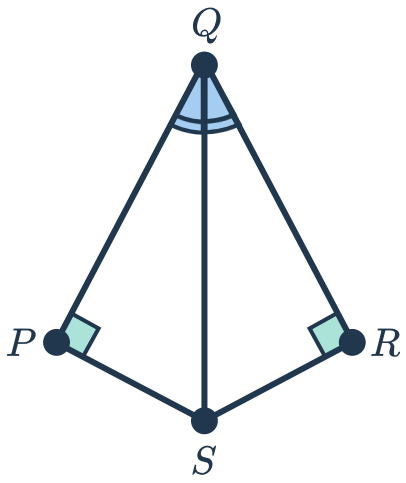


© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 Construct a proof of the angle bisector theorem.
Given:

- $\overline{QS}$ bisects $\angle PQR$
- $\overline{SP} \perp \overline{PQ}$
- $\overline{SR} \perp \overline{RQ}$

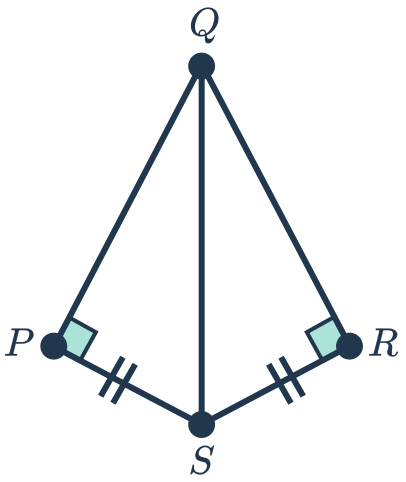
Prove: $SP = SR$



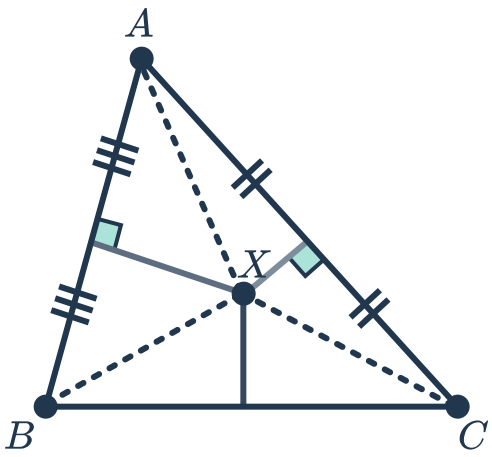
11 Construct a proof of the converse of angle bisector theorem.
Given:

- $SP = SR$
- $\overline{SP} \perp \overline{PQ}$
- $\overline{SR} \perp \overline{RQ}$

Prove: $m\angle PQS = m\angle RQS$

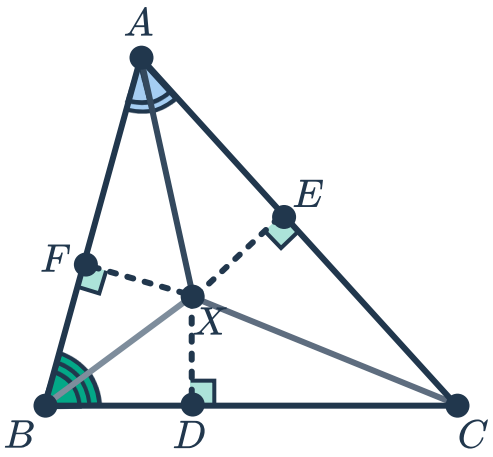


12 Given that the perpendicular bisectors of $\overline{AB}$ and $\overline{AC}$ intersect at the point X, prove that X lies on the perpendicular bisector of $\overline{BC}$.



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

13 Given that the angle bisectors of $\angle ABC$ and $\angle BAC$ intersect at the point X, prove that X lies on the angle bisector of $\angle ACB$.



© ClipSAT: By Mr. Mohamed Abdallah +966597688647